

MICROSOFT ACCESS

Complete Visual Study Manual

EDA-ACC-105

Beginner to Intermediate

6 Weeks | 18 Lessons

Professional Certificate in Microsoft Access

Classroom + Guided Database Practicals + Final Database Project

ETHAN DIGITAL ACADEMY

How to Use This Manual

This manual is written as a complete learner study book. Read each lesson, study the visual examples, complete the guided practical using fictional practice data, answer the review questions and complete the assignment before moving on.

Course Learning Outcomes

- Explain relational database concepts and the purpose of Microsoft Access.
- Create well-structured tables with suitable fields, data types and primary keys.
- Build relationships and maintain data integrity.
- Retrieve information using filters and select queries.
- Create practical forms for data entry and reports for professional output.
- Import, export, back up and maintain a small Access database.
- Plan, build, test and present a complete relational database project.

Course Map

1. Introduction to Databases and Microsoft Access
2. Access Interface and Database Objects
3. Planning a Relational Database
4. Creating a Database and Tables
5. Fields, Data Types and Field Properties
6. Primary Keys and Data Integrity
7. Entering, Editing and Sorting Records
8. Filtering and Finding Data
9. Relationships Between Tables
10. Creating Select Queries
11. Query Criteria and Calculated Fields
12. Creating Forms
13. Improving Forms and Data Entry
14. Creating Reports
15. Grouping, Sorting and Summaries in Reports
16. Importing and Exporting Data
17. Database Maintenance, Backup and Good Practice
18. Final Relational Database Project

Lesson 1: Introduction to Databases and Microsoft Access

Introduction

A database is an organised collection of related information. Microsoft Access is a relational database application that helps users store, connect, retrieve and present structured records. Unlike a simple list, a relational database can separate information into logical tables and connect those tables through keys.

Learning Objectives

- Explain the main ideas in introduction to databases and microsoft access.
- Apply the lesson using a safe fictional Ethan Digital Academy dataset.
- Check the accuracy of the result and explain why the method is useful.

Key Terms

Database, Relational Database, Record, Field, Table, Database Management System.

Teaching Content

Microsoft Access works best when information is organised deliberately. A learner should always ask what each table represents, what each field means, and how the information will be used later. Clear structure is more important than entering a large quantity of data.

Professional database work depends on consistency. Names, dates, codes and money values should follow agreed rules. When the structure is consistent, queries and reports become more reliable and users can understand the database more easily.

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Database vs Spreadsheet

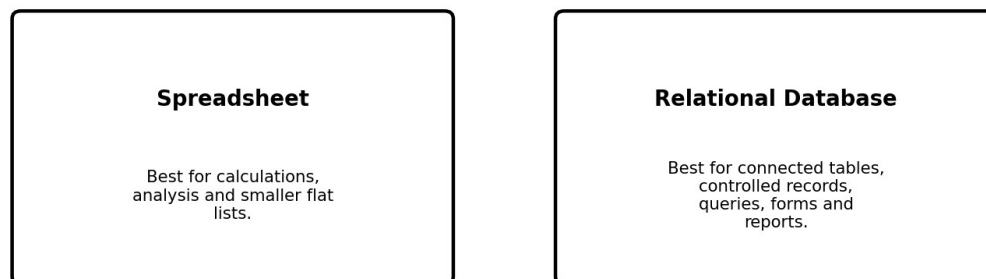


Figure 1. Instructional visual for Introduction to Databases and Microsoft Access.

Lesson 1 Practical Workshop

Step-by-Step Practical

Compare a course register in a spreadsheet with a database that separates learners, courses and enrolments. Identify repeated information and explain why repetition can create errors.

1. Prepare a practice copy of the database and use only fictional learner/business information.
2. Open the relevant Access object and confirm you are in the correct view before editing.
3. Carry out the task carefully, using meaningful names for fields and objects.
4. Save the object, close it, reopen it and verify that the change was stored correctly.
5. Test the result with at least three fictional records and note anything that needs correction.

Ethan Digital Academy Example

Imagine Ethan Digital Academy needs a reliable way to manage courses and enrolments. Apply this lesson to a fictional academy database. The aim is to reduce repeated information, improve accuracy and make useful information easier to retrieve.

Review Questions

- What is the main purpose of introduction to databases and microsoft access?
- Which Access object or concept is most important in this lesson, and why?
- What error could occur if this lesson is applied carelessly?
- How would you verify that your work is correct?

Assignment

Sketch three tables for an academy database: Learners, Courses and Enrolments.

Lesson 2: Access Interface and Database Objects

Introduction

An Access database is built from objects. Tables store data, queries retrieve or calculate information, forms support controlled data entry, and reports present information professionally. The Navigation Pane helps users open and organise these objects. Ribbon commands change according to the active object and view.

Learning Objectives

- Explain the main ideas in access interface and database objects.
- Apply the lesson using a safe fictional Ethan Digital Academy dataset.
- Check the accuracy of the result and explain why the method is useful.

Key Terms

Navigation Pane, Ribbon, Datasheet View, Design View, Tables, Queries, Forms, Reports.

Teaching Content

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Microsoft Access: Main Database Objects



Figure 2. Instructional visual for Access Interface and Database Objects.

Lesson 2 Practical Workshop

Step-by-Step Practical

Open a blank database and identify the Ribbon, Navigation Pane, object tabs and status area. Create one empty table and switch between Datasheet View and Design View.

1. Prepare a practice copy of the database and use only fictional learner/business information.
2. Open the relevant Access object and confirm you are in the correct view before editing.
3. Carry out the task carefully, using meaningful names for fields and objects.
4. Save the object, close it, reopen it and verify that the change was stored correctly.
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Ethan Digital Academy Example

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Review Questions

- What is the main purpose of access interface and database objects?
- Which Access object or concept is most important in this lesson, and why?
- What error could occur if this lesson is applied carelessly?
- How would you verify that your work is correct?

Assignment

Draw and label the four main Access objects and state one use for each.

Lesson 3: Planning a Relational Database

Introduction

Good databases begin with planning. Start from the business questions the database must answer, identify entities, list their attributes, choose unique identifiers and decide how entities relate. Avoid putting unrelated facts in one huge table. A clear plan reduces duplication and makes future queries easier.

Learning Objectives

- Explain the main ideas in planning a relational database.
- Apply the lesson using a safe fictional Ethan Digital Academy dataset.
- Check the accuracy of the result and explain why the method is useful.

Key Terms

Entity, Attribute, Relationship, Normalization Concept, Business Rule.

Teaching Content

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Plan Before You Build

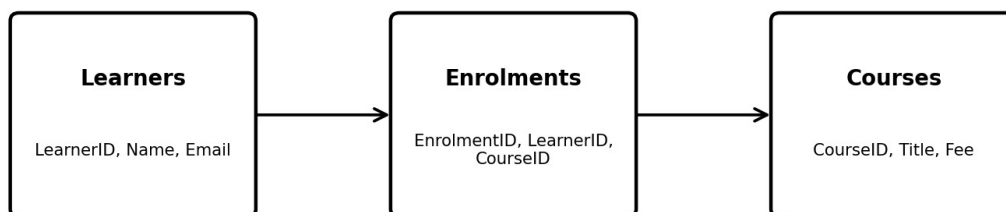


Figure 3. Instructional visual for Planning a Relational Database.

Lesson 3 Practical Workshop

Step-by-Step Practical

For Ethan Digital Academy, list the information needed to answer: Who enrolled? Which course? When? What fee applies? Divide the information into appropriate tables.

1. Prepare a practice copy of the database and use only fictional learner/business information.
2. Open the relevant Access object and confirm you are in the correct view before editing.
3. Carry out the task carefully, using meaningful names for fields and objects.
4. Save the object, close it, reopen it and verify that the change was stored correctly.
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Ethan Digital Academy Example

Imagine Ethan Digital Academy needs a reliable way to manage courses and enrolments. Apply this lesson to a fictional academy database. The aim is to reduce repeated information, improve accuracy and make useful information easier to retrieve.

Review Questions

- What is the main purpose of planning a relational database?
- Which Access object or concept is most important in this lesson, and why?
- What error could occur if this lesson is applied carelessly?
- How would you verify that your work is correct?

Assignment

Prepare a simple data dictionary for Learners, Courses and Enrolments.

Lesson 4: Creating a Database and Tables

Introduction

A new Access file normally uses the .accdb format. Tables can be created in Datasheet View for quick entry or Design View for precise control. Meaningful table and field names make the database easier to maintain. Each table should represent one main subject.

Learning Objectives

- Explain the main ideas in creating a database and tables.
- Apply the lesson using a safe fictional Ethan Digital Academy dataset.
- Check the accuracy of the result and explain why the method is useful.

Key Terms

Accdb, Table, Field Name, Design View, Datasheet View.

Teaching Content

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Table Anatomy

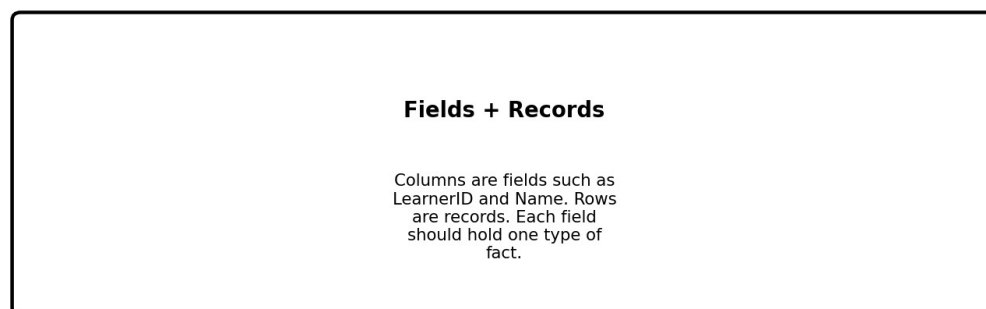


Figure 4. Instructional visual for Creating a Database and Tables.

Lesson 4 Practical Workshop

Step-by-Step Practical

Create a database named EDA_Training.accdb. Create tblLearners with LearnerID, FullName, Email and Phone fields. Save and reopen the table.

1. Prepare a practice copy of the database and use only fictional learner/business information.
2. Open the relevant Access object and confirm you are in the correct view before editing.
3. Carry out the task carefully, using meaningful names for fields and objects.
4. Save the object, close it, reopen it and verify that the change was stored correctly.
5. Test the result with at least three fictional records and note anything that needs correction.

Ethan Digital Academy Example

Imagine Ethan Digital Academy needs a reliable way to manage courses and enrolments. Apply this lesson to a fictional academy database. The aim is to reduce repeated information, improve accuracy and make useful information easier to retrieve.

Review Questions

- What is the main purpose of creating a database and tables?
- Which Access object or concept is most important in this lesson, and why?
- What error could occur if this lesson is applied carelessly?
- How would you verify that your work is correct?

Assignment

Create tblCourses with CourseID, CourseTitle, DurationWeeks and Fee.

Lesson 5: Fields, Data Types and Field Properties

Introduction

Every field needs a suitable data type. Short Text stores names and codes, Number supports numeric calculations, Date/Time stores dates, Currency stores money, Yes/No stores two-state values, and Long Text supports longer notes. Field properties such as Required, Field Size, Format and Default Value improve consistency.

Learning Objectives

- Explain the main ideas in fields, data types and field properties.
- Apply the lesson using a safe fictional Ethan Digital Academy dataset.
- Check the accuracy of the result and explain why the method is useful.

Key Terms

Data Type, Field Property, Required, Default Value, Validation.

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Choosing Suitable Data Types

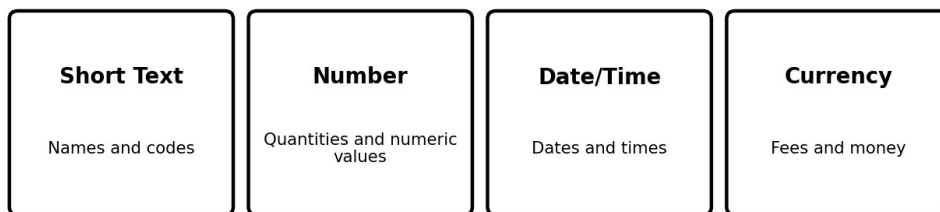


Figure 5. Instructional visual for Fields, Data Types and Field Properties.

Lesson 5 Practical Workshop

Step-by-Step Practical

In tblCourses, set CourseTitle to Short Text, DurationWeeks to Number and Fee to Currency. Mark CourseTitle as required. Enter three fictional courses and review how each value is displayed.

1. Prepare a practice copy of the database and use only fictional learner/business information.
2. Open the relevant Access object and confirm you are in the correct view before editing.
3. Carry out the task carefully, using meaningful names for fields and objects.
4. Save the object, close it, reopen it and verify that the change was stored correctly.
5. Test the result with at least three fictional records and note anything that needs correction.

Ethan Digital Academy Example

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Review Questions

- What is the main purpose of fields, data types and field properties?
- Which Access object or concept is most important in this lesson, and why?
- What error could occur if this lesson is applied carelessly?
- How would you verify that your work is correct?

Assignment

Explain why a phone number is usually better stored as Short Text than Number.

Lesson 6: Primary Keys and Data Integrity

Introduction

A primary key uniquely identifies each record. AutoNumber is convenient for generated identifiers, while meaningful codes can also be used when uniqueness is guaranteed. Primary keys prevent ambiguity and support relationships. Data integrity means keeping records accurate, valid and consistent.

Learning Objectives

- Explain the main ideas in primary keys and data integrity.
- Apply the lesson using a safe fictional Ethan Digital Academy dataset.
- Check the accuracy of the result and explain why the method is useful.

Key Terms

Primary Key, Unique Value, Autonumber, Duplicate, Integrity.

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Primary Key = Unique Identity

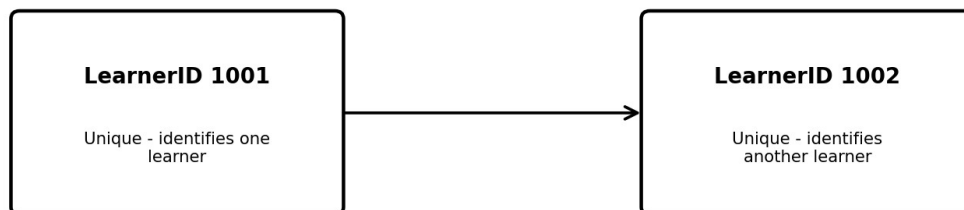


Figure 6. Instructional visual for Primary Keys and Data Integrity.

Lesson 6 Practical Workshop

Step-by-Step Practical

Set LearnerID as an AutoNumber primary key. Enter two fictional learners and observe the automatically generated IDs. Try to understand why names should not normally be primary keys.

1. Prepare a practice copy of the database and use only fictional learner/business information.
2. Open the relevant Access object and confirm you are in the correct view before editing.
3. Carry out the task carefully, using meaningful names for fields and objects.
4. Save the object, close it, reopen it and verify that the change was stored correctly.
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Ethan Digital Academy Example

Imagine Ethan Digital Academy needs a reliable way to manage courses and enrolments. Apply this lesson to a fictional academy database. The aim is to reduce repeated information, improve accuracy and make useful information easier to retrieve.

Review Questions

- What is the main purpose of primary keys and data integrity?
- Which Access object or concept is most important in this lesson, and why?
- What error could occur if this lesson is applied carelessly?
- How would you verify that your work is correct?

Assignment

Write four rules that improve data integrity in a learner database.

Lesson 7: Entering, Editing and Sorting Records

Introduction

Datasheet View allows records to be entered and edited in rows. Users should verify values before leaving a record, avoid duplicate entries and use consistent formats. Sorting changes the display order without changing the underlying meaning of the data.

Learning Objectives

- Explain the main ideas in entering, editing and sorting records.
- Apply the lesson using a safe fictional Ethan Digital Academy dataset.
- Check the accuracy of the result and explain why the method is useful.

Key Terms

Record Selector, Sort Ascending, Sort Descending, Edit, Delete.

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Lesson 7 Practical Workshop

Step-by-Step Practical

Enter five fictional learner records. Sort FullName A-Z, then sort CourseTitle in a course table. Correct one deliberately mistyped fictional value.

1. Prepare a practice copy of the database and use only fictional learner/business information.
2. Open the relevant Access object and confirm you are in the correct view before editing.
3. Carry out the task carefully, using meaningful names for fields and objects.
4. Save the object, close it, reopen it and verify that the change was stored correctly.
5. Test the result with at least three fictional records and note anything that needs correction.

Ethan Digital Academy Example

Imagine Ethan Digital Academy needs a reliable way to manage courses and enrolments. Apply this lesson to a fictional academy database. The aim is to reduce repeated information, improve accuracy and make useful information easier to retrieve.

Review Questions

- What is the main purpose of entering, editing and sorting records?
- Which Access object or concept is most important in this lesson, and why?
- What error could occur if this lesson is applied carelessly?
- How would you verify that your work is correct?

Assignment

Create a one-page data-entry checklist for staff.

Lesson 8: Filtering and Finding Data

Introduction

Filtering temporarily displays records that match a condition. Find locates specific values. These tools are useful for quick review, but complex or reusable questions are better handled with queries. Filters do not delete records that are hidden from view.

Learning Objectives

- Explain the main ideas in filtering and finding data.
- Apply the lesson using a safe fictional Ethan Digital Academy dataset.
- Check the accuracy of the result and explain why the method is useful.

Key Terms

Filter, Selection Filter, Text Filter, Find, Replace.

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Lesson 8 Practical Workshop

Step-by-Step Practical

Filter fictional courses to show only those with fees above a chosen amount. Clear the filter. Use Find to locate one fictional learner by name.

1. Prepare a practice copy of the database and use only fictional learner/business information.
2. Open the relevant Access object and confirm you are in the correct view before editing.
3. Carry out the task carefully, using meaningful names for fields and objects.
4. Save the object, close it, reopen it and verify that the change was stored correctly.
5. Test the result with at least three fictional records and note anything that needs correction.

Ethan Digital Academy Example

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Review Questions

- What is the main purpose of filtering and finding data?
- Which Access object or concept is most important in this lesson, and why?
- What error could occur if this lesson is applied carelessly?
- How would you verify that your work is correct?

Assignment

Describe two situations where a filter is enough and two where a query is better.

Lesson 9: Relationships Between Tables

Introduction

Relationships connect tables using matching key fields. A common one-to-many relationship means one record in a parent table can relate to many records in another table. Referential integrity helps prevent orphaned references and protects consistency.

Learning Objectives

- Explain the main ideas in relationships between tables.
- Apply the lesson using a safe fictional Ethan Digital Academy dataset.
- Check the accuracy of the result and explain why the method is useful.

Key Terms

Foreign Key, One-To-Many, Referential Integrity, Parent Table, Related Table.

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One-to-Many Relationship

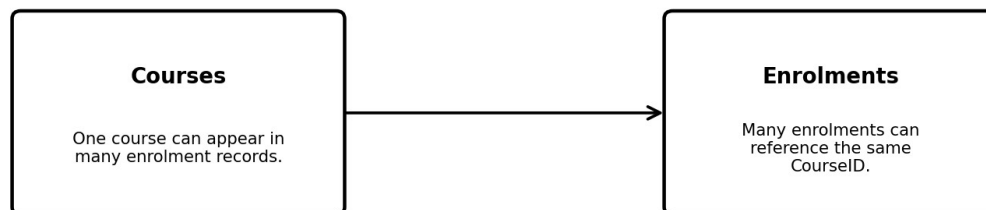


Figure 7. Instructional visual for Relationships Between Tables.

Lesson 9 Practical Workshop

Step-by-Step Practical

Open Relationships. Add Learners, Courses and Enrolments. Connect LearnerID and CourseID to their matching foreign-key fields in Enrolments. Review the one-to-many symbols.

1. Prepare a practice copy of the database and use only fictional learner/business information.
2. Open the relevant Access object and confirm you are in the correct view before editing.
3. Carry out the task carefully, using meaningful names for fields and objects.
4. Save the object, close it, reopen it and verify that the change was stored correctly.
5. Test the result with at least three fictional records and note anything that needs correction.

Ethan Digital Academy Example

Imagine Ethan Digital Academy needs a reliable way to manage courses and enrolments. Apply this lesson to a fictional academy database. The aim is to reduce repeated information, improve accuracy and make useful information easier to retrieve.

Review Questions

- What is the main purpose of relationships between tables?
- Which Access object or concept is most important in this lesson, and why?
- What error could occur if this lesson is applied carelessly?
- How would you verify that your work is correct?

Assignment

Draw the relationship diagram and explain each connection in one sentence.

Lesson 10: Creating Select Queries

Introduction

A select query retrieves chosen fields from one or more tables without changing the original records. Query Design lets users add tables, choose fields, set sorting and run the query. Queries turn stored data into useful information.

Learning Objectives

- Explain the main ideas in creating select queries.
- Apply the lesson using a safe fictional Ethan Digital Academy dataset.
- Check the accuracy of the result and explain why the method is useful.

Key Terms

Select Query, Query Design, Field Grid, Run, Result Set.

Teaching Content

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Query Design Workflow

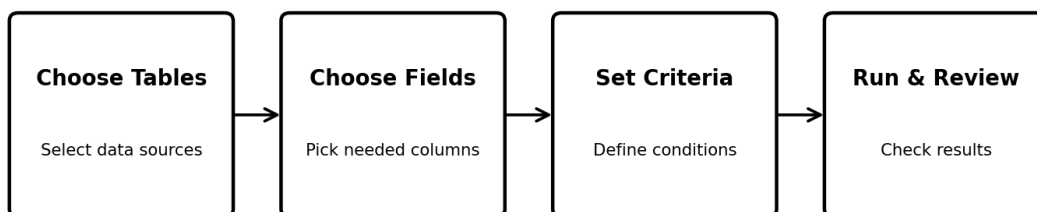


Figure 8. Instructional visual for Creating Select Queries.

Lesson 10 Practical Workshop

Step-by-Step Practical

Create qryLearnerCourses showing learner name, course title and enrolment date from related tables. Run it and verify that each row makes sense.

1. Prepare a practice copy of the database and use only fictional learner/business information.
2. Open the relevant Access object and confirm you are in the correct view before editing.
3. Carry out the task carefully, using meaningful names for fields and objects.
4. Save the object, close it, reopen it and verify that the change was stored correctly.
5. Test the result with at least three fictional records and note anything that needs correction.

Ethan Digital Academy Example

Imagine Ethan Digital Academy needs a reliable way to manage courses and enrolments. Apply this lesson to a fictional academy database. The aim is to reduce repeated information, improve accuracy and make useful information easier to retrieve.

Review Questions

- What is the main purpose of creating select queries?
- Which Access object or concept is most important in this lesson, and why?
- What error could occur if this lesson is applied carelessly?
- How would you verify that your work is correct?

Assignment

Create a query listing all fictional courses with title, duration and fee.

Lesson 11: Query Criteria and Calculated Fields

Introduction

Criteria restrict query results. Calculated fields derive a new value from existing data, such as Balance: [Fee]-[AmountPaid]. Criteria must match the field data type and should be tested carefully so valid records are not accidentally excluded.

Learning Objectives

- Explain the main ideas in query criteria and calculated fields.
- Apply the lesson using a safe fictional Ethan Digital Academy dataset.
- Check the accuracy of the result and explain why the method is useful.

Key Terms

Criteria, Expression, Calculated Field, Comparison Operator, Parameter Concept.

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Query Criteria Examples

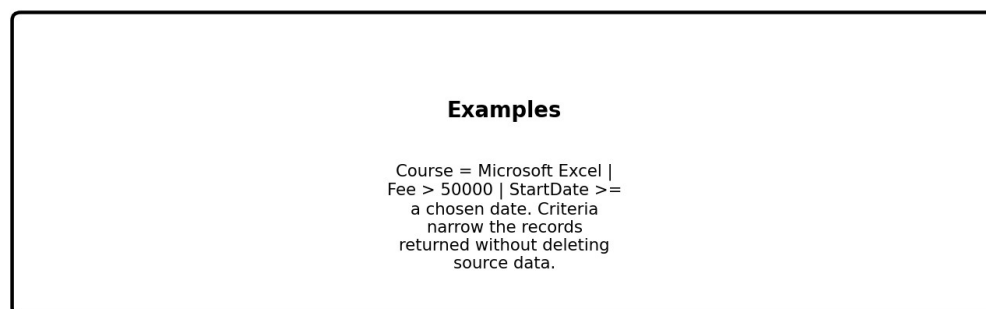


Figure 9. Instructional visual for Query Criteria and Calculated Fields.

Lesson 11 Practical Workshop

Step-by-Step Practical

Create a query showing courses whose Fee is greater than a chosen threshold. Add a simple calculated field using fictional Fee and AmountPaid values.

1. Prepare a practice copy of the database and use only fictional learner/business information.
2. Open the relevant Access object and confirm you are in the correct view before editing.
3. Carry out the task carefully, using meaningful names for fields and objects.
4. Save the object, close it, reopen it and verify that the change was stored correctly.
5. Test the result with at least three fictional records and note anything that needs correction.

Ethan Digital Academy Example

Imagine Ethan Digital Academy needs a reliable way to manage courses and enrolments. Apply this lesson to a fictional academy database. The aim is to reduce repeated information, improve accuracy and make useful information easier to retrieve.

Review Questions

- What is the main purpose of query criteria and calculated fields?
- Which Access object or concept is most important in this lesson, and why?
- What error could occur if this lesson is applied carelessly?
- How would you verify that your work is correct?

Assignment

Write five example criteria for text, number and date fields.

Lesson 12: Creating Forms

Introduction

Forms provide a controlled, user-friendly way to view or enter records. A form can be generated from a table or query and then improved in Layout or Design View. Clear labels and logical tab order reduce data-entry mistakes.

Learning Objectives

- Explain the main ideas in creating forms.
- Apply the lesson using a safe fictional Ethan Digital Academy dataset.
- Check the accuracy of the result and explain why the method is useful.

Key Terms

Form, Control, Label, Text Box, Record Source, Form View.

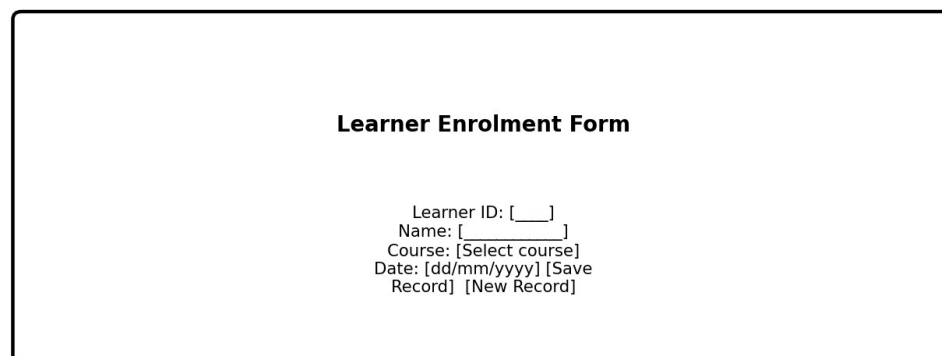
Teaching Content

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Form Design



Learner Enrolment Form

Learner ID: []
Name: []
Course: [Select course]
Date: [dd/mm/yyyy] [Save
Record] [New Record]

Figure 10. Instructional visual for Creating Forms.

Lesson 12 Practical Workshop

Step-by-Step Practical

Create a Learner form from tblLearners. Add one fictional record through the form rather than directly in the table. Confirm that the record appears in the table.

1. Prepare a practice copy of the database and use only fictional learner/business information.
2. Open the relevant Access object and confirm you are in the correct view before editing.
3. Carry out the task carefully, using meaningful names for fields and objects.
4. Save the object, close it, reopen it and verify that the change was stored correctly.
5. Test the result with at least three fictional records and note anything that needs correction.

Ethan Digital Academy Example

Imagine Ethan Digital Academy needs a reliable way to manage courses and enrolments. Apply this lesson to a fictional academy database. The aim is to reduce repeated information, improve accuracy and make useful information easier to retrieve.

Review Questions

- What is the main purpose of creating forms?
- Which Access object or concept is most important in this lesson, and why?
- What error could occur if this lesson is applied carelessly?
- How would you verify that your work is correct?

Assignment

List six qualities of a professional data-entry form.

Lesson 13: Improving Forms and Data Entry

Introduction

Useful forms guide the user. Controls can be resized and aligned; combo boxes can offer approved choices; validation can prevent unacceptable values; and headings can clarify purpose. Design should support accuracy rather than decoration.

Learning Objectives

- Explain the main ideas in improving forms and data entry.
- Apply the lesson using a safe fictional Ethan Digital Academy dataset.
- Check the accuracy of the result and explain why the method is useful.

Key Terms

Combo Box, Control Source, Tab Order, Validation Rule, Layout.

Teaching Content

Microsoft Access works best when information is organised deliberately. A learner should always ask what each table represents, what each field means, and how the information will be used later. Clear structure is more important than entering a large quantity of data.

Professional database work depends on consistency. Names, dates, codes and money values should follow agreed rules. When the structure is consistent, queries and reports become more reliable and users can understand the database more easily.

A database object should have a clear purpose. Before creating or changing an object, identify the question it solves. After the change, test the object with fictional records and confirm that the output matches the intended business rule.

Lesson 13 Practical Workshop

Step-by-Step Practical

Improve the learner or enrolment form: align controls, add a clear title and use a suitable list control for course selection if available. Test the form with fictional data.

1. Prepare a practice copy of the database and use only fictional learner/business information.
2. Open the relevant Access object and confirm you are in the correct view before editing.
3. Carry out the task carefully, using meaningful names for fields and objects.
4. Save the object, close it, reopen it and verify that the change was stored correctly.
5. Test the result with at least three fictional records and note anything that needs correction.

Ethan Digital Academy Example

Imagine Ethan Digital Academy needs a reliable way to manage courses and enrolments. Apply this lesson to a fictional academy database. The aim is to reduce repeated information, improve accuracy and make useful information easier to retrieve.

Review Questions

- What is the main purpose of improving forms and data entry?
- Which Access object or concept is most important in this lesson, and why?
- What error could occur if this lesson is applied carelessly?
- How would you verify that your work is correct?

Assignment

Sketch a professional enrolment form and label its controls.

Lesson 14: Creating Reports

Introduction

Reports format data for viewing, printing or PDF output. A report may be based on a table or, preferably, a query that already selects the required records. Headers, detail sections and footers organise the output.

Learning Objectives

- Explain the main ideas in creating reports.
- Apply the lesson using a safe fictional Ethan Digital Academy dataset.
- Check the accuracy of the result and explain why the method is useful.

Key Terms

Report, Report Wizard, Report Header, Detail, Page Footer.

Teaching Content

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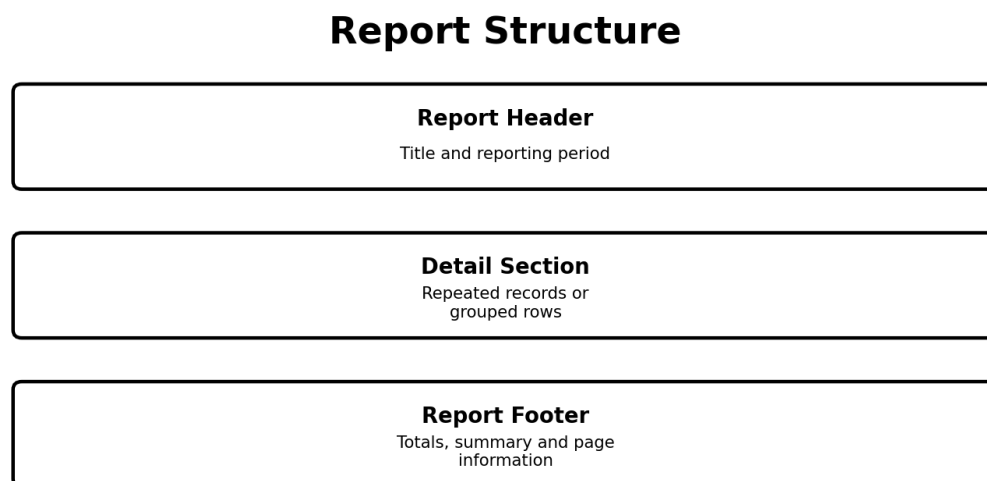


Figure 11. Instructional visual for Creating Reports.

Lesson 14 Practical Workshop

Step-by-Step Practical

Create a report from qryLearnerCourses. Add a title such as Ethan Digital Academy Enrolment Report. Preview the report and check that columns fit.

1. Prepare a practice copy of the database and use only fictional learner/business information.
2. Open the relevant Access object and confirm you are in the correct view before editing.
3. Carry out the task carefully, using meaningful names for fields and objects.
4. Save the object, close it, reopen it and verify that the change was stored correctly.
5. Test the result with at least three fictional records and note anything that needs correction.

Ethan Digital Academy Example

Imagine Ethan Digital Academy needs a reliable way to manage courses and enrolments. Apply this lesson to a fictional academy database. The aim is to reduce repeated information, improve accuracy and make useful information easier to retrieve.

Review Questions

- What is the main purpose of creating reports?
- Which Access object or concept is most important in this lesson, and why?
- What error could occur if this lesson is applied carelessly?
- How would you verify that your work is correct?

Assignment

Explain why a query can be a better Record Source for a report than a raw table.

Lesson 15: Grouping, Sorting and Summaries in Reports

Introduction

Reports become more informative when records are grouped by a meaningful field such as CourseTitle. Sorting determines order within groups, while totals can summarize counts or numeric values. Every summary should have a clear label and purpose.

Learning Objectives

- Explain the main ideas in grouping, sorting and summaries in reports.
- Apply the lesson using a safe fictional Ethan Digital Academy dataset.
- Check the accuracy of the result and explain why the method is useful.

Key Terms

Grouping, Sorting, Aggregate, Count, Sum, Group Footer.

Teaching Content

Microsoft Access works best when information is organised deliberately. A learner should always ask what each table represents, what each field means, and how the information will be used later. Clear structure is more important than entering a large quantity of data.

Professional database work depends on consistency. Names, dates, codes and money values should follow agreed rules. When the structure is consistent, queries and reports become more reliable and users can understand the database more easily.

A database object should have a clear purpose. Before creating or changing an object, identify the question it solves. After the change, test the object with fictional records and confirm that the output matches the intended business rule.

Lesson 15 Practical Workshop

Step-by-Step Practical

Group a fictional enrolment report by course. Add a count of enrolments for each course and inspect the report in Print Preview.

1. Prepare a practice copy of the database and use only fictional learner/business information.
2. Open the relevant Access object and confirm you are in the correct view before editing.
3. Carry out the task carefully, using meaningful names for fields and objects.
4. Save the object, close it, reopen it and verify that the change was stored correctly.
5. Test the result with at least three fictional records and note anything that needs correction.

Ethan Digital Academy Example

Imagine Ethan Digital Academy needs a reliable way to manage courses and enrolments. Apply this lesson to a fictional academy database. The aim is to reduce repeated information, improve accuracy and make useful information easier to retrieve.

Review Questions

- What is the main purpose of grouping, sorting and summaries in reports?
- Which Access object or concept is most important in this lesson, and why?
- What error could occur if this lesson is applied carelessly?
- How would you verify that your work is correct?

Assignment

Design a report specification showing title, grouping, sort order and required totals.

Lesson 16: Importing and Exporting Data

Introduction

Access can exchange data with formats such as Excel, text/CSV and PDF. Importing brings external data into Access, while exporting sends Access data to another format. Before import, inspect field names and data types. After export, verify that the result is complete and readable.

Learning Objectives

- Explain the main ideas in importing and exporting data.
- Apply the lesson using a safe fictional Ethan Digital Academy dataset.
- Check the accuracy of the result and explain why the method is useful.

Key Terms

Import, Export, Csv, Excel, Pdf, External Data.

Teaching Content

Microsoft Access works best when information is organised deliberately. A learner should always ask what each table represents, what each field means, and how the information will be used later. Clear structure is more important than entering a large quantity of data.

Professional database work depends on consistency. Names, dates, codes and money values should follow agreed rules. When the structure is consistent, queries and reports become more reliable and users can understand the database more easily.

A database object should have a clear purpose. Before creating or changing an object, identify the question it solves. After the change, test the object with fictional records and confirm that the output matches the intended business rule.

Lesson 16 Practical Workshop

Step-by-Step Practical

Create a small fictional course list in a spreadsheet or CSV and import it into a practice table. Export a query or report to PDF and verify the output.

1. Prepare a practice copy of the database and use only fictional learner/business information.
2. Open the relevant Access object and confirm you are in the correct view before editing.
3. Carry out the task carefully, using meaningful names for fields and objects.
4. Save the object, close it, reopen it and verify that the change was stored correctly.
5. Test the result with at least three fictional records and note anything that needs correction.

Ethan Digital Academy Example

Imagine Ethan Digital Academy needs a reliable way to manage courses and enrolments. Apply this lesson to a fictional academy database. The aim is to reduce repeated information, improve accuracy and make useful information easier to retrieve.

Review Questions

- What is the main purpose of importing and exporting data?
- Which Access object or concept is most important in this lesson, and why?
- What error could occur if this lesson is applied carelessly?
- How would you verify that your work is correct?

Assignment

Write a pre-import checklist with at least six checks.

Lesson 17: Database Maintenance, Backup and Good Practice

Introduction

Databases need care. Keep backup copies, use clear naming, avoid storing unnecessary personal data, restrict access appropriately and document important design decisions. Compact and Repair can help maintain an Access file. Changes to structure should be tested on a backup first.

Learning Objectives

- Explain the main ideas in database maintenance, backup and good practice.
- Apply the lesson using a safe fictional Ethan Digital Academy dataset.
- Check the accuracy of the result and explain why the method is useful.

Key Terms

Backup, Compact And Repair, Documentation, Access Control, Data Minimisation.

Teaching Content

Microsoft Access works best when information is organised deliberately. A learner should always ask what each table represents, what each field means, and how the information will be used later. Clear structure is more important than entering a large quantity of data.

Professional database work depends on consistency. Names, dates, codes and money values should follow agreed rules. When the structure is consistent, queries and reports become more reliable and users can understand the database more easily.

A database object should have a clear purpose. Before creating or changing an object, identify the question it solves. After the change, test the object with fictional records and confirm that the output matches the intended business rule.

Lesson 17 Practical Workshop

Step-by-Step Practical

Make a copy of the practice database before a structural change. Review object names and create a short README note describing the tables and relationships.

1. Prepare a practice copy of the database and use only fictional learner/business information.
2. Open the relevant Access object and confirm you are in the correct view before editing.
3. Carry out the task carefully, using meaningful names for fields and objects.
4. Save the object, close it, reopen it and verify that the change was stored correctly.
5. Test the result with at least three fictional records and note anything that needs correction.

Ethan Digital Academy Example

Imagine Ethan Digital Academy needs a reliable way to manage courses and enrolments. Apply this lesson to a fictional academy database. The aim is to reduce repeated information, improve accuracy and make useful information easier to retrieve.

Review Questions

- What is the main purpose of database maintenance, backup and good practice?
- Which Access object or concept is most important in this lesson, and why?
- What error could occur if this lesson is applied carelessly?
- How would you verify that your work is correct?

Assignment

Prepare a database maintenance schedule for a small training academy.

Lesson 18: Final Relational Database Project

Introduction

The final project combines planning, tables, keys, relationships, queries, forms and reports. The goal is not merely to create objects but to build a small database that answers real business questions accurately and is easy for another user to operate.

Learning Objectives

- Explain the main ideas in final relational database project.
- Apply the lesson using a safe fictional Ethan Digital Academy dataset.
- Check the accuracy of the result and explain why the method is useful.

Key Terms

Requirements, Database Design, Testing, Documentation, Presentation.

Teaching Content

Microsoft Access works best when information is organised deliberately. A learner should always ask what each table represents, what each field means, and how the information will be used later. Clear structure is more important than entering a large quantity of data.

Professional database work depends on consistency. Names, dates, codes and money values should follow agreed rules. When the structure is consistent, queries and reports become more reliable and users can understand the database more easily.

A database object should have a clear purpose. Before creating or changing an object, identify the question it solves. After the change, test the object with fictional records and confirm that the output matches the intended business rule.

Final Project Architecture

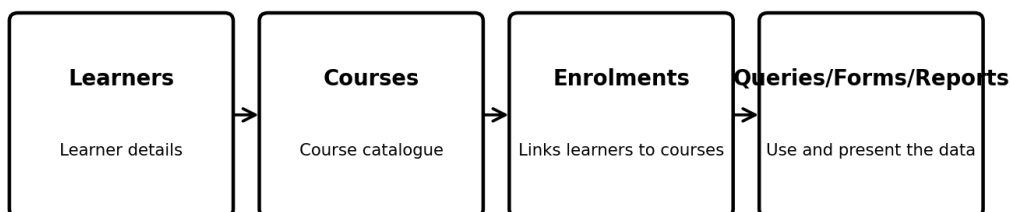


Figure 12. Instructional visual for Final Relational Database Project.

Lesson 18 Practical Workshop

Step-by-Step Practical

Build the Ethan Digital Academy Training Database using fictional data: at least three related tables, valid primary keys, relationships, three useful queries, two forms and two reports. Test every object and correct errors.

1. Prepare a practice copy of the database and use only fictional learner/business information.
2. Open the relevant Access object and confirm you are in the correct view before editing.
3. Carry out the task carefully, using meaningful names for fields and objects.
4. Save the object, close it, reopen it and verify that the change was stored correctly.
5. Test the result with at least three fictional records and note anything that needs correction.

Ethan Digital Academy Example

Imagine Ethan Digital Academy needs a reliable way to manage courses and enrolments. Apply this lesson to a fictional academy database. The aim is to reduce repeated information, improve accuracy and make useful information easier to retrieve.

Review Questions

- What is the main purpose of final relational database project?
- Which Access object or concept is most important in this lesson, and why?
- What error could occur if this lesson is applied carelessly?
- How would you verify that your work is correct?

Assignment

Submit the database, a one-page design explanation, test evidence and a short presentation of how the system works.

Final Practical Project - 100 Marks

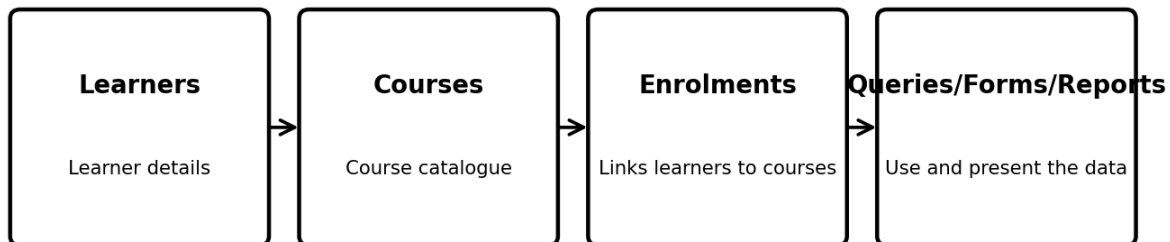
Build a complete fictional Ethan Digital Academy Training Database. Use no real personal data.

- Database plan and data dictionary - 10 marks
- Three or more correctly designed tables - 20 marks
- Primary keys and relationships with integrity - 15 marks
- Three useful queries including criteria/calculation - 15 marks
- Two professional data-entry forms - 10 marks
- Two professional reports with grouping/summary - 10 marks
- Testing, backup and documentation - 10 marks
- Presentation and explanation - 10 marks

Required Business Questions

- Which learners are enrolled in each course?
- How many enrolments does each course have?
- Which courses meet a selected fee criterion?
- What information should appear on a management enrolment report?

Final Project Architecture



Course Quiz - 30 Questions

1. What is a database?
2. What is a relational database?
3. What is a table?
4. What is a field?
5. What is a record?
6. Name the four main Access objects.
7. What is the purpose of a primary key?
8. What is a foreign key?
9. What does one-to-many mean?
10. What is referential integrity?
11. When is Short Text appropriate?
12. When is Currency appropriate?
13. What does Required do?
14. What is Datasheet View?
15. What is Design View?
16. What is a filter?
17. What is a select query?
18. What does query criteria do?
19. What is a calculated field?
20. What is a form?
21. Why use a combo box?
22. What is a report?
23. What is grouping in a report?
24. What is an aggregate such as Count?
25. What does importing data mean?
26. What does exporting data mean?
27. Why should imported data be checked?
28. Why are backups important?
29. What is Compact and Repair used for?
30. Why should a database be tested before production use?

Final Examination - 100 Marks

Section A - Concepts (30 Marks)

- Define database, table, field, record, primary key and foreign key. (12)
- Explain the purpose of tables, queries, forms and reports. (8)
- Explain one-to-many relationships and referential integrity. (10)

Section B - Applied Database Design (30 Marks)

- Design a three-table academy database and identify fields/data types. (15)
- Draw the relationships and identify primary/foreign keys. (10)
- Write five useful query questions the database should answer. (5)

Section C - Practical (40 Marks)

- Create the tables and relationships. (15)
- Create one query with criteria and one calculated field. (10)
- Create one form. (5)
- Create one grouped report. (5)
- Back up and document the database. (5)

Course Completion Outcome

After completing EDA-ACC-105, the learner should be able to plan and build a small relational Microsoft Access database, create and relate tables, enter and retrieve records accurately, design queries, forms and reports, exchange data with common formats, maintain backups and explain the design decisions used in a practical database project.

Professional Practice Checklist

- Use meaningful object and field names.
- Choose data types deliberately.
- Use primary keys and relationships correctly.
- Use fictional practice data during training.
- Validate and test before relying on results.
- Back up before structural changes.
- Protect personal and business information appropriately.
- Document the database so another authorised user can understand it.